

Automated Medical Podcast Transcription and Topic Segmentation

1. Introduction

In recent years, medical podcasts have emerged as a valuable source of information for medical students, healthcare professionals, researchers, and educators. These podcasts cover a wide range of topics including diseases, symptoms, treatment methods, clinical experiences, case discussions, and public health awareness. Due to their conversational and explanatory nature, podcasts help listeners gain deeper insights into real-world medical scenarios.

Medical podcasts are also increasingly used as supplementary learning resources alongside textbooks and journals. Experts often share real-life experiences and updated medical practices, making podcasts an effective medium for continuous medical education.

However, most medical podcasts are lengthy and unstructured. Identifying a specific disease discussion or treatment explanation within a long audio episode requires listening to the entire podcast, which is both time-consuming and inefficient. This creates a major challenge for users who wish to quickly access relevant medical information.

To address this issue, there is a strong need for an intelligent AI-based system that can automatically convert medical podcast audio into text, divide it into meaningful topic-based segments, and provide structured insights such as keywords, sentiment analysis, and transcription quality evaluation.

The **Automated Medical Podcast Transcription and Topic Segmentation** project aims to solve this problem using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques. The system transforms raw medical audio into searchable, structured, and user-friendly content, enabling efficient access and analysis of medical podcast information.

2. Problem Statement

Medical podcasts contain highly valuable educational and clinical information, but their long duration and lack of structure make them difficult to analyze and navigate. Users face several challenges, such as locating specific disease or treatment discussions, understanding topic transitions, and evaluating the quality and reliability of the content.

Since podcasts are audio-based, traditional search and indexing methods cannot be applied directly. This makes it difficult for users to reuse podcast content for academic or reference purposes.

Currently, there is no effective automated mechanism that can process medical podcast audio, extract meaningful topics, and provide analytical insights in a structured format. As a result, users are forced to rely on manual listening and interpretation.

The problem addressed by this project is the design and development of an intelligent system that can automatically transcribe medical podcast audio, detect topic boundaries within the transcript, extract keywords, analyze sentiment, evaluate transcription quality, and present the results through an interactive user interface.

3. Objectives of the Project

The primary objective of this project is to develop an AI-driven system that improves the accessibility and usability of medical podcast content. The system aims to accurately convert medical audio into text using advanced speech-to-text models and organize the information in a structured manner.

The project also aims to segment long transcripts into meaningful medical topics, making it easier for users to navigate large podcast episodes. Another objective is to extract important medical keywords that summarize each topic segment.

In addition, the system performs sentiment analysis in a medical-aware manner and evaluates transcription quality using metrics such as Word Error Rate (WER), Character Error Rate (CER), and semantic similarity. The final objective is to provide an interactive frontend interface for visualization and analysis.

4. Scope of the Project

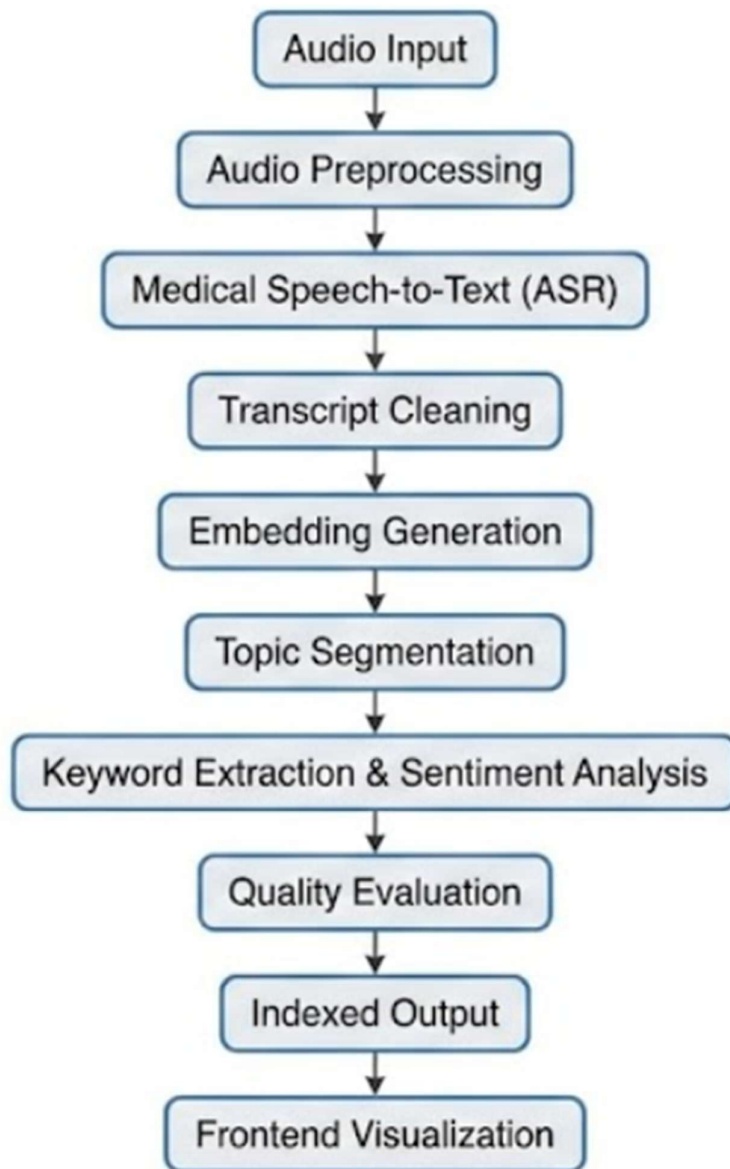
The scope of this project includes the processing of recorded medical podcast audio files. It focuses on automatic transcription using Automatic Speech Recognition (ASR) models, topic segmentation using NLP techniques, keyword extraction, sentiment analysis, and transcription quality evaluation using standard metrics.

The project also includes visualization of results through a web-based interface for better user interaction.

The project does not include live audio streaming or real-time transcription. It also does not provide clinical diagnosis or treatment recommendations, as the system is intended strictly for educational and informational purposes.

5. System Architecture :

The system uses a pipeline-based modular architecture to process medical podcast audio. The audio is transcribed using speech-to-text models and analyzed using NLP techniques for topic segmentation, keyword extraction, sentiment analysis, and quality evaluation. The results are stored and visualized through an interactive frontend interface.



6. Module Description

6.1 Audio Preprocessing Module

The audio preprocessing module prepares raw medical podcast audio for accurate transcription. Medical podcasts often contain background noise, silence, and varying volume levels. This module normalizes the audio volume, removes unnecessary silence, and reduces noise using audio processing libraries. The cleaned audio improves transcription accuracy.

6.2 Transcription Module

The transcription module converts preprocessed audio into text using advanced speech-to-text models such as Whisper or Faster-Whisper. These models handle conversational speech and medical terminology effectively. Timestamps are generated for spoken segments, enabling precise alignment between audio and text. The transcript is stored in structured JSON format.

6.3 Topic Segmentation Module

The topic segmentation module divides long transcripts into smaller, meaningful topic-based segments. Sentence embeddings are generated to capture semantic meaning. Similarity between consecutive sentences is computed to detect topic change boundaries, resulting in topic-wise segmented transcripts.

6.4 Keyword Extraction Module

This module extracts important medical terms from each topic segment using NLP-based algorithms such as KeyBERT, YAKE, and RAKE. Extracted keywords help users quickly understand the main subject of each segment.

6.5 Sentiment and Quality Evaluation Module

Sentiment analysis is performed at the segment level to classify discussions as positive, neutral, or negative. Medical-aware normalization ensures informative discussions are not misclassified.

For quality evaluation, generated transcripts are compared with clean reference transcripts using WER, CER, and semantic similarity. These metrics help assess transcription reliability.

6.6 Frontend Visualization Module

The frontend module presents processed data through an interactive web interface. It displays transcripts, topic summaries, keywords, sentiment graphs, and quality dashboards. Timestamp-based navigation improves usability.

7. Technology Stack

Backend: Python 3.9+, Flask, Whisper / Faster-Whisper, Librosa, PyDub, FFmpeg

NLP & Machine Learning: HuggingFace Transformers, Sentence Transformers, NLTK, SpaCy, KeyBERT, YAKE, RAKE

Frontend: React.js, HTML, CSS, JavaScript, REST APIs

Visualization: Chart.js, Plotly, Matplotlib

Storage: JSON, CSV, SQLite (optional), FAISS (optional)

8. Project Structure

8.1 Backend Structure

File	Description
transcription.py	Converts audio into text and generates timestamps
segmentation.py	Detects topic boundaries using embeddings
evaluation_summary.py	Computes WER, CER, accuracy, similarity
sentiment_analysis.py	Performs medical-aware sentiment analysis
keyword_cloud.py	Extracts and counts medical keywords
logger.py	Tracks pipeline execution and errors

8.2 API Layer

The API layer is implemented using Flask. The app.py file handles audio uploads, triggers backend processing modules, and sends results to the frontend via REST APIs.

8.3 Frontend Structure

The frontend provides dashboards and visual analytics for transcripts, topics, sentiment, and quality metrics. Timestamp-based navigation allows efficient access to specific sections.

8.4 Data and Output Folders

Separate folders are maintained for clean reference transcripts, generated transcripts, and output files such as JSON and CSV. This structure ensures organized data management.

9. Milestone-wise Implementation

The project was implemented in phases starting with audio preprocessing and transcription, followed by topic segmentation and keyword extraction. Sentiment analysis and quality evaluation were implemented next, and frontend integration and visualization were completed later, followed by documentation and final submission.

10. Data and Privacy Considerations

Audio files are processed temporarily and are not permanently stored. API keys are handled securely using environment variables. No personal user data is stored.

11. Challenges Faced

Some challenges included handling medical terminology accurately, managing background noise in audio, and tuning topic segmentation thresholds. Ensuring correct sentiment classification for medical discussions was also a challenge.

12. Limitations

The system supports only recorded audio files and does not handle real-time transcription. Accuracy may vary depending on audio quality and speaker clarity. The system does not provide clinical diagnosis.

13. Troubleshooting

Low transcription accuracy can be improved by enhancing audio quality. Topic segmentation issues can be resolved by adjusting similarity thresholds. Errors are tracked using the logging module for debugging.

14. Future Scope

Future enhancements include Medical Named Entity Recognition, speaker diarization, multilingual transcription support, semantic search, and automatic summarization of topics.

15. Intended Users

Medical students, healthcare professionals, medical researchers, and medical educators.

16. Conclusion

The Automated Medical Podcast Transcription and Topic Segmentation project provides an AI-based solution for analyzing long medical podcasts. By integrating speech recognition, NLP, sentiment analysis, quality evaluation, and visualization, the system makes medical audio content structured, searchable, and accessible.

17. License

This project is licensed under the **MIT License**.

18. References

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 2. HuggingFace Transformers Documentation
 3. Sentence Transformers Research Papers
 4. Librosa Audio Processing Library Documentation
 5. Research studies on ASR and topic segmentation
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